

4. Insights and Recommendations Report

4.1 Key Insights

- **Overtime Work is the Strongest Predictor:** Employees working overtime exhibit significantly higher attrition propensity.
- **Monthly Compensation & Job Level Impact:** Lower monthly income and lower job levels strongly correlate with increased employee turnover.
- **Tenure and Experience Dynamics:** Employees with lower total working years, fewer years at the current company, and shorter tenure under their current manager show a higher probability of leaving.
- **Age & Demographic Factors:** Younger employees and single employees demonstrate higher churn rates compared to older or married staff.
- **Commute & Work Distance:** Greater distance from home increases dissatisfaction and likelihood of default/departure.

4.2 Feature Importance

Top contributing features driving employee attrition based on feature evaluation and model metrics:

1. **MonthlyIncome** (Highest impact on retention stability)
2. **Age**
3. **OverTime**
4. **DailyRate / MonthlyRate**
5. **TotalWorkingYears**
6. **DistanceFromHome**
7. **YearsAtCompany**
8. **YearsWithCurrManager**

4.3 Business Recommendations

1. Workload & Overtime Management

- **Review Overtime Policy:** Monitor workloads for high-risk departments (e.g., Sales, Laboratory) and implement strict overtime limits to prevent burnout.

- **Resource Allocation:** Reallocate tasks or hire contract support in bottleneck areas to reduce mandatory overtime hours.

2. Competitive Compensation & Benchmark Reviews

- **Address Pay Disparities:** Target salary adjustments and competitive hike structures for entry-to-mid level roles, especially those falling significantly below department average monthly income.
- **Stock Options & Incentives:** Implement targeted stock option plans and performance bonuses for early-career hires to improve long-term retention.

3. Career Pathing & Leadership Engagement

- **Early Tenure Engagement:** Establish structured 30-60-90 day onboarding and mentorship programs for new hires and early-career personnel.
- **Managerial Alignment:** Train direct managers on empathetic leadership and regular check-ins, targeting employees in their first 1-3 years under a new manager.